PHENOBARBITAL

Respiratory depression
Controlled drug

CHEMICAL NAME 5-ethyl-5-phenylpyrimidine-2,4,6(1H,3H,5H)-trione (phenobarbitone)
CLASS Barbiturate antiepileptic
TRADE NAMES Luminal · Gardenal · Phenobarbitone sodium injection

THERAPEUTIC USES / INDICATIONS

- Neonatal seizures (first-line).
- Status epilepticus (second/third-line).
- All seizure types except absence (low-resource settings).
- Alcohol/benzodiazepine withdrawal (some protocols).

ADMINISTRATION

- **IV** slowly (diluted 1:10 with water for injection) max 1 mg/kg/min (≤100 mg/min) with airway support.
- **PO** at bedtime (sedation).
- **IM** deep (avoid intra-arterial).

DOSAGES

- **Loading (seizures/status):** 15–20 mg/kg IV slowly with monitoring (US label: children/infants).
- **Maintenance (children):** 3–6 mg/kg/day (UK: <20 kg 5 mg/kg/day).
- **Maintenance (adult):** 60–200 mg/day (UK: 2–3 mg/kg at bedtime).
- **Levels:** use the local lab reference range (loading gives about 20 mcg/mL).

PHARMACOKINETICS

- **Onset:** IV 5–20 min; PO 1 h. **Duration:** 10–12 h+.
- Metabolised in liver (CYP2C9, 2C19); potent enzyme inducer.
- Excreted in urine (↑ with alkaline urine). **t½:** 75–120 h (adults); longer in neonates.

PHARMACODYNAMICS

- Prolongs opening of **GABA-A chloride channels** (and directly opens at high doses); ↓ glutamate effects → anticonvulsant, sedative.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Sedation, drowsiness.
- Hyperactivity, irritability (children); cognitive impairment.
- Ataxia, nystagmus.

ADVERSE EFFECTS (SERIOUS)

- Respiratory depression, hypotension (IV).
- Dependence; withdrawal seizures.
- SJS; megaloblastic anaemia, osteomalacia.
- Teratogenic (cardiac malformations); neonatal withdrawal.

CONTRAINDICATIONS & CAUTIONS

- Acute porphyria.
- Severe respiratory depression; severe hepatic impairment.
- **Caution:** elderly, depression, pregnancy.

KEY INTERACTIONS

- Enzyme inducer: ↓ oral contraceptives, warfarin, steroids, many drugs.
- **Alcohol, opioids, benzodiazepines:** additive CNS/respiratory depression.
- **Valproate:** ↑ phenobarbital levels.

NURSING MANAGEMENT

- IV: monitor RR, SpO_2, BP; airway equipment ready.
- Monitor levels, sedation, seizure frequency.
- Neonates: monitor breathing and feeding.
- Taper slowly when stopping.

PATIENT EDUCATION

- Causes drowsiness; no driving or alcohol.
- Never stop suddenly.
- Hormonal contraception unreliable.
- Store safely; controlled medicine.

KEY POINTS

- Barbiturate — GABA-A potentiator.
- Respiratory depression, sedation.
- Neonatal seizures first-line; status 20 mg/kg IV.
- Very long t½; enzyme inducer.

SOURCES CHECKED **[739]** Phenobarbital Colonis 50 mg/5 ml Oral Suspension (UK SmPC) **[740]** Phenobarbital, oral (US label) **[741]** Phenobarbital, oral (US label, American Health Packaging)

Full references p. 432–457